CHURNGUIDE: BUSINESS-ADAPTABLE CUSTOMER CHURN PREDICTION

Project Report submitted

in partial fulfillment of the requirements for the degree

of

[Degree and Programme]

by

[Student name(s) and enrolment number(s)]

Under the guidance of

[Supervisor name]

[Designation]

[Institution logo]

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ACKNOWLEDGMENTS

The public dataset and published technical sources used in this report are identified in the references. The project uses the Python scientific-computing ecosystem and the libraries listed in the implementation chapter. Their documentation supports the description of the methods and the distinction between predictive associations and causal claims.

[Candidate-specific acknowledgments to the supervisor, department and other contributors are to be completed by the candidate after confirming their names and contributions.]

[Name(s) of the candidate(s)]

ABSTRACT

Customer churn analysis helps organisations review customers whose historical characteristics are associated with cancellation or non-renewal. This project presents ChurnGuide, a local web application for business-adaptable customer churn prediction from labelled tabular snapshots. A three-step workflow supports data selection, column mapping and result review. Project-specific preprocessing and classifiers are saved together, allowing different customer schemas to use the same interface without treating one trained model as universal.

Numeric and categorical transformations are fitted inside a machine-learning pipeline. Logistic regression, random forest and gradient boosting are compared through stratified five-fold cross-validation on the training partition of an eighty-twenty split. The selected model is then evaluated on the held-out partition.

The public telecommunications demonstration contains 7,043 records and selects gradient boosting, with a held-out receiver-operating-characteristic area of 0.8465 and recall of 0.5000. A generated software-account fixture contains 1,200 records and demonstrates a second schema; its results are not evidence from a real enterprise.

The application provides individual and batch scoring, historical customer analytics, numeric segmentation and model diagnostics. Individual explanations use baseline perturbations, including a linked telecom billing calculation, and are interpreted as model sensitivity rather than causal effects. Seventeen of eighteen included software tests passed; the remaining assertion expects the former home-page redirect. Saved-model predictions matched the recorded held-out metrics.

The project demonstrates an integrated analytical prototype. Prospective temporal validation, probability calibration and controlled retention studies remain necessary before operational effectiveness can be established.

TABLE OF CONTENTS

DESCRIPTION	PAGE NUMBER
Certificate	iii
Declaration	v
Acknowledgments	vii
Abstract	ix
List of figures	xv
List of tables	xvii
Abbreviations	xix
Chapter 1 INTRODUCTION	1
1.1 Background And Motivation	1
1.2 Problem Statement	2
1.3 Objectives And Scope	3
1.4 Implementation Contribution	4
Chapter 2 LITERATURE REVIEW	5
2.1 Churn Prediction Research	5
2.2 Classification Methods	6
2.3 Preprocessing And Evaluation Practice	7
2.4 Interpretation And Research Gap	8
Chapter 3 REQUIREMENTS AND DATA	9
3.1 Functional Requirements	9

TABLE OF CONTENTS (CONTINUED)

DESCRIPTION	PAGE NUMBER
3.2 Data Contract And Validation	10
3.3 Public Telecommunications Demonstration	11
3.4 Generated Software-Account Demonstration	12
Chapter 4 SYSTEM DESIGN AND METHODOLOGY	13
4.1 System Architecture	13
4.2 Project Artifact Design	14
4.3 Preprocessing Pipeline	15
4.4 Partitioning And Model Selection	16
4.5 Classification Metrics	17
4.6 Global And Local Model Interpretation	18
4.7 Customer Segmentation Method	19
4.8 Decision Support And Responsible Use	20
Chapter 5 IMPLEMENTATION	21
5.1 Implementation Environment	21
5.2 Wizard: Data And Business Profile	22
5.3 Wizard: Column Review And Results	23
5.4 Individual Customer Scoring	24
5.5 Batch Scoring And Customer Analytics	25
5.6 Model And Segment Views	26
Chapter 6 RESULTS AND DISCUSSION	27
6.1 Evaluation Evidence And Reproducibility	27
6.2 Telecommunications Model Comparison	28
6.3 Telecommunications Held-Out Results	29
6.4 Generated Software Demonstration Results	30
6.5 Feature And Segment Interpretation	31

TABLE OF CONTENTS (CONTINUED)

DESCRIPTION	PAGE NUMBER
6.6 Software Validation And Discussion	32
Chapter 7 CONCLUSION AND FUTURE WORK	33
7.1 Conclusions	33
7.2 Future Work	34
REFERENCES	35
APPENDIX A DATA DICTIONARIES	37
A.1 Telecommunications Feature Dictionary	37
A.2 Generated Software Feature Dictionary	38
APPENDIX B REPRODUCIBILITY AND USER WORKFLOW	39
B.1 Reproducibility Checklist	39
B.2 User Workflow And Interpretation Checklist	40

LIST OF FIGURES

FIGURE	TITLE	PAGE NUMBER
1.1	Project workflow from data to customer review	4
4.1	Logical architecture of the local application	13
4.2	Preprocessing and classification pipeline	15
4.3	Exploratory segmentation workflow	19
5.1	Data-selection and business-profile screen from the supplied application	22
5.2	Software-account individual-scoring screen; the underlying demo data is synthetic	24
5.3	Held-out telecommunications confusion matrix saved by the supplied project	26
6.1	Saved telecommunications held-out receiver-operating-characteristic curve	28

LIST OF TABLES

TABLE	TITLE	PAGE NUMBER
1.1	Problem definition and analytical boundaries	2
1.2	Project objectives and implementation outputs	3
2.1	Candidate models and practical trade-offs	6
2.2	Different evaluation questions	7
3.1	Functional requirements	9
3.2	Training-data checks in the main validation function	10
3.3	Telecommunications demonstration schema summary	11
3.4	Generated software demonstration schema summary	12
4.1	Main software responsibilities	13
4.2	Project artifacts and their analytical role	14
4.3	Implemented hyperparameter search grids	16
4.4	Definitions used for classification metrics	17
4.5	Interpretation methods and boundaries	18
4.6	Operational questions before real deployment	20
5.1	Runtime package versions in the supplied requirements	21
5.2	Main wizard stages and expected user decisions	23
5.3	Display thresholds by scoring view	25
6.1	Evidence available for this report	27
6.2	Recorded five-fold training-partition comparison for telecommunications	28
6.3	Telecommunications held-out classification results	29
6.4	Recorded training comparison on the synthetic software fixture	30
6.5	Synthetic software held-out results	30
6.6	Top saved permutation diagnostics: mean decrease in held-out area	31

LIST OF TABLES (CONTINUED)

TABLE	TITLE	PAGE NUMBER
6.7	Saved telecommunications exploratory groups	31
6.8	Validation categories and interpretation	32
7.1	Future work and the evidence it would add	34
A.1	Telecommunications project feature roles	37
A.2	Synthetic software project feature roles	38
B.1	Reproduction steps and expected evidence	39
B.2	Reading the main outputs	40

ABBREVIATIONS / NOTATIONS

AUC Area under the receiver operating characteristic curve

CSV Comma-separated values

CV Cross-validation

F1 Harmonic mean of precision and recall

FN / FP False negative / false positive

GB Gradient boosting

ID Identifier

LR Logistic regression

RF Random forest

ROC Receiver operating characteristic

SaaS Software as a service

TN / TP True negative / true positive

p Estimated positive-class probability

CHAPTER 1
INTRODUCTION

1.1 Background And Motivation

Customer churn is the loss of an existing customer relationship through cancellation, non-renewal or another business-defined exit. The operational meaning changes across organisations. A telecommunications provider may record service disconnection, whereas a software provider may record a cancelled account. A useful analytical system must therefore connect the outcome label to the business context before it interprets a prediction.

Customer records often combine numerical measures, such as account age and spending, with categories, such as payment method and subscription plan. These mixed records can support historical analysis and supervised classification. However, a list of predictions alone does not explain how the data was prepared, how a model was selected or whether the outcome definition is appropriate. These questions motivate a single workflow that brings data preparation, model comparison and customer review together.

ChurnGuide is a local web application built around business-specific tabular customer snapshots. It provides a three-step wizard, individual and batch scoring, descriptive analytics, customer segments and model diagnostics. The supplied implementation includes two separate demonstration projects. The telecommunications example uses a public customer dataset, while the business-to-business software example uses generated data to demonstrate a different schema. Neither example establishes that one trained model transfers unchanged to every business.

1.2 Problem Statement

The project addresses the difficulty of turning a labelled customer spreadsheet into an inspectable churn analysis without hard-coding every business column into the interface. The input is a table containing one historical customer snapshot per row, a binary outcome and selected predictors. The required output is a project-specific model and a set of views that help a user inspect estimated risk and historical patterns.

Table 1.1 Problem definition and analytical boundaries

Element	Definition
Unit of analysis	A customer/account snapshot in a tabular file
Outcome	A user-mapped binary churn label
Predictors	Selected numeric and categorical attributes
Primary output	Estimated positive-class probability
Supporting outputs	Risk bands, diagnostics, analytics and segments
Excluded claim	Guaranteed retention or causal intervention effect

Table 1.1 separates the modelling task from the business decision that follows it. A high model score can prioritise a case for review, but it does not establish that the customer will leave or that an offer will prevent departure. The system supports analysis rather than an automatic customer-contact or pricing decision.

1.3 Objectives And Scope

The first objective is to accept different customer schemas through an explicit mapping of the identifier, outcome, positive outcome value and usable features. The second is to keep preprocessing inside the learning pipeline so that imputation and scaling are fitted within the training process. The third is to compare candidate classifiers using training-partition cross-validation, followed by an evaluation of the selected model on a held-out partition.

Table 1.2 Project objectives and implementation outputs

Objective	Implementation output
Flexible setup	Upload, inspect and map a customer table
Repeatable modelling	Saved pipeline and experiment metadata
Customer review	Single-record probability and sensitivity
Portfolio review	Batch scoring with export and risk bands
Historical understanding	Categorical rates and numeric summaries
Group understanding	Numeric customer segments and profiles

The scope is business-adaptable binary classification, not unrestricted automation for every spreadsheet. The current workflow expects labelled training data, selected tabular predictors and an outcome with exactly two valid classes. Free-text modelling, time-to-event estimation, streaming deployment and experiment-based retention optimisation are outside the evaluated scope.

1.4 Implementation Contribution

The implementation contribution is the integration of schema mapping, project-specific model artifacts and customer-facing analysis in one local application. A project stores its own configuration, cleaned dataset, trained classifier, evaluation metadata and plots. The same interface can then display a telecommunications example with nineteen predictors or a generated software-account example with eleven predictors. This demonstrates software adaptability while keeping each model tied to its own schema.

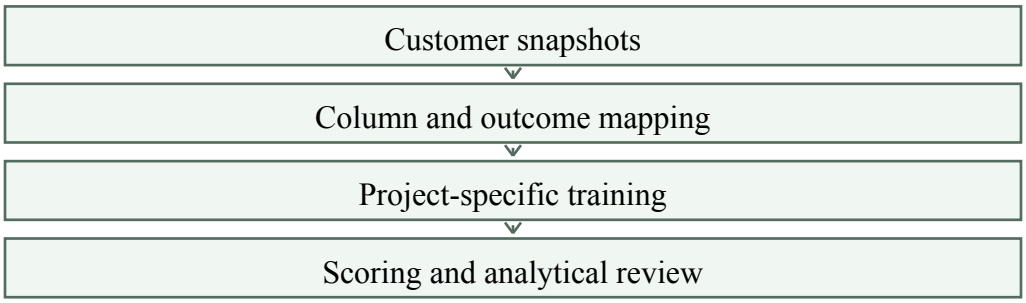


Figure 1.1 Project workflow from data to customer review

CHAPTER 2
LITERATURE REVIEW

2.1 Churn Prediction Research

Churn prediction studies usually connect a particular outcome to a particular organisation, dataset and observation period. Ahmad, Jafar and Aljoumaa [1] studied telecommunications churn using an industrial dataset and a big-data processing environment. Their work includes domain-specific feature construction and social-network information. It supports the importance of understanding the source data, but its experimental performance cannot be treated as a benchmark directly comparable with the smaller public snapshot used in this project.

The present application differs in emphasis. It packages a moderate-scale tabular workflow for local use and makes the schema configurable. It does not implement the distributed data infrastructure or social-network feature engineering in [1]. Its public telecommunications demonstration and generated software demonstration answer narrower questions: whether the modelling pipeline works with the supplied columns, and whether the software can operate with different business schemas.

This distinction matters when discussing generality. A reusable training procedure can be applied to another labelled table, but its predictive usefulness must be assessed again. Differences in customer populations, label definitions, billing systems and available features can change performance. Cross-business software flexibility is therefore reported separately from predictive validation on a real business population.

2.2 Classification Methods

Three classifier families are implemented. Logistic regression provides a regularised linear decision function after feature transformation. It is a useful starting point when the data may be described by additive effects on log-odds. Random forest averages many decision trees and can represent nonlinear interactions. Breiman [2] describes the random-forest approach as an ensemble of tree predictors built using randomisation.

Gradient boosting builds an additive model in successive stages. Each stage contributes a correction to the existing model according to the optimisation objective. Friedman [3] develops the general gradient-boosting framework. The supplied application uses the scikit-learn implementation of gradient-boosted classification trees rather than an external boosting package.

Table 2.1 Candidate models and practical trade-offs

Method	Strength	Consideration
Logistic regression	Compact regularised baseline	Linear log-odds assumption
Random forest	Nonlinear interactions	Tree depth and ensemble size
Gradient boosting	Sequential nonlinear fitting	Learning rate and tree complexity

Table 2.1 summarises modelling choices, not a claim that one family is always superior. The application compares the candidate families within the same split and scoring protocol. The observed winner is specific to each project.

2.3 Preprocessing And Evaluation Practice

Pedregosa et al. [5] describe scikit-learn as a machine-learning library with a consistent interface for supervised and unsupervised methods. That interface allows transformations and a classifier to be composed as a pipeline. The official guidance on common pitfalls [7] explains why preprocessing fitted before the train-test split can expose information from evaluation data to the learning process.

The application follows this principle for statistical preprocessing: the numeric imputer, standard scaler, categorical imputer and encoder are pipeline components. Cross-validation operates on the entire pipeline, so each training fold learns its own transformation statistics. This arrangement reduces a specific source of leakage. It does not prove that a selected feature was measured before the outcome or that a random split represents a future operating period.

Table 2.2 Different evaluation questions

Question	Relevant evidence
Can the software fit and score?	Unit tests and application checks
Does ranking work on held-out rows?	Held-out receiver-operating-characteristic area
Will future customers be ranked well?	Time-ordered prospective evaluation
Will outreach reduce churn?	Controlled intervention study

The evidence categories in Table 2.2 must remain distinct. Passing software tests is not the same as proving business effectiveness, and a probability in the range zero to one is not evidence of calibration.

2.4 Interpretation And Research Gap

Model interpretation can be global or local. Permutation importance evaluates how a fitted model relies on a feature by shuffling that feature and measuring the change in a chosen score. The official documentation [8] notes that the result is specific to the fitted model and that correlated features complicate interpretation. In this project, permutation importance is a held-out diagnostic, not a basis for a second round of feature selection using the same test set.

For individual customers, ChurnGuide replaces one selected feature at a time with a typical training value and measures the probability change. This is local model sensitivity, not a tree-path decomposition, a Shapley-value explanation or a causal effect. The distinction is important because correlated customer attributes can make a one-feature replacement an unusual combination of values.

The practical gap addressed by the application is the separation that often exists between data preparation, classifier evaluation and customer review screens. The project joins these activities within a schema-driven local tool. Its contribution is the implemented workflow and transparent separation of evidence types. It does not claim a new learning algorithm or universal churn performance. Numeric clustering follows the standard distance-based approach described in [6] and is used only as exploratory segmentation.

CHAPTER 3
REQUIREMENTS AND DATA

3.1 Functional Requirements

The intended user is an analyst or business user reviewing a customer table on a local computer. A training workflow begins with a labelled file and a description of what churn means for that business. The main wizard exposes data selection, column review and results as separate stages. Subsequent screens support the inspection of one customer, a scoring file or the historical customer portfolio.

Table 3.1 Functional requirements

ID	Requirement
FR1	Accept a customer CSV and profile its columns
FR2	Map identifier, binary target and positive value
FR3	Select numeric and categorical predictors
FR4	Train and save a project-specific pipeline
FR5	Display comparison and held-out diagnostics
FR6	Estimate individual and batch customer risk
FR7	Display analytics and numeric customer segments
FR8	Provide a scored customer export

Table 3.1 states the core workflow requirements. It should not be read as an exhaustive acceptance certificate for every input or every deployment configuration. Evaluation evidence is reported in Chapter 6 with its tested scope.

3.2 Data Contract And Validation

The training contract requires a binary target with a user-selected positive value. The validator checks for at least fifty rows and at least ten examples of each class. It excludes the mapped target and identifier from the predictor list and requires at least two selected predictors. These are implementation rules supporting this workflow, not universal statistical guarantees about minimum sample size.

Table 3.2 Training-data checks in the main validation function

Check	Purpose
Two target values	Defines binary classification
Positive value exists	Controls which outcome is coded as one
Duplicate selected IDs	Flags repeated non-missing identifiers
Nonempty varying features	Avoids unusable selected predictors
Categorical cardinality at most 100	Limits one-hot feature expansion
Known feature columns	Ensures selected columns exist

Numeric cleaning strips surrounding whitespace, converts values with pandas and treats failed conversions or infinities as missing. Categorical cleaning strips surrounding whitespace and replaces selected empty markers. Missing predictor values are subsequently imputed inside the learning pipeline. These operations standardise representation; they do not establish the business validity of an observation. A data owner must still check units, dates, outcome timing and whether a selected column reveals the answer indirectly.

3.3 Public Telecommunications Demonstration

The supplied telecommunications file contains 7,043 customer records and twenty-one columns. The identifier is customerID and the target is Churn, with Yes mapped to the positive class. The model uses three numeric and sixteen categorical predictors. IBM's public repository [4] is the stated source for the sample. It is not presented as a private client dataset.

Table 3.3 Telecommunications demonstration schema summary

Role	Columns or count
Identifier	customerID
Target	Churn; positive value Yes
Numeric features	tenure, MonthlyCharges, TotalCharges
Categorical features	16 service, account and demographic fields
Outcome distribution	1,869 positive and 5,174 negative records
Missing-value example	11 whitespace entries in TotalCharges

Table 3.3 describes the supplied snapshot. A configured outcome-window text does not create missing timestamp evidence. The report therefore treats this dataset as cross-sectional historical classification rather than a verified thirty-day prospective forecast. The 80/20 stratified split contains 5,634 training rows and 1,409 held-out rows. The statistical pipeline imputes missing numeric charges using the training median, rather than filling the entire source file before partitioning.

3.4 Generated Software-Account Demonstration

The second demonstration contains 1,200 generated business-to-business software accounts. The supplied script creates this fixture to exercise a different schema. The identifier is Account_ID, the target is Status and Churned is the positive value. There are six numeric and five categorical predictors. This dataset is synthetic and must not be interpreted as evidence from a deployed software company.

Table 3.4 Generated software demonstration schema summary

Role	Columns or count
Identifier	Account_ID
Target	Status; positive value Churned
Numeric features	Account age, licences, spend, activity, tickets, adoption
Categorical features	Plan, company size, cycle, payment, auto-renew
Outcome counts	499 positive and 701 negative records
Partition sizes	960 training and 240 held-out rows

The two examples demonstrate different identifiers, target names and feature sets. They do not establish successful deployment across all industries. New applications require a new labelled dataset, a defensible churn definition and their own validation. Generated data is useful for interface and schema testing because its provenance is known, but any relationships it contains reflect its generation procedure rather than observed enterprise behaviour.

CHAPTER 4
SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

A browser presents server-rendered forms and result views. Flask routes call the data, project, training, explanation and segmentation modules. Project directories hold the datasets and fitted artifacts. This separates learning code from page templates and lets project configuration determine the interface fields.

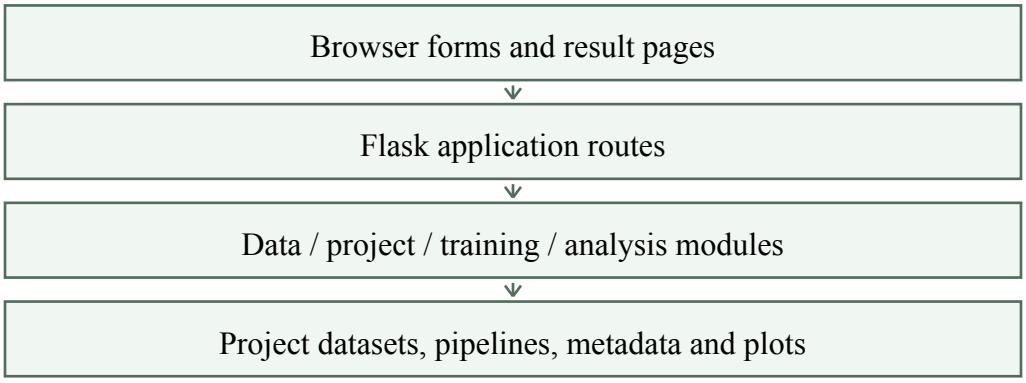


Figure 4.1 Logical architecture of the local application

Table 4.1 Main software responsibilities

Module	Responsibility
app.py	Routes, form handling and page context
data_loader.py	Schema profile, mapping validation and typing
project_manager.py	Project artifacts and scoring access
train_experiment.py	Training search and held-out diagnostics
explain.py	Local sensitivity and review heuristics
segmentation.py	Numeric cluster summaries

4.2 Project Artifact Design

Each project directory stores config.json, dataset.csv, pipeline.joblib, metadata.json, segmentation.json and a plots directory. The configuration describes the business context and selected feature roles. The saved pipeline contains fitted preprocessing and the classifier. Metadata records the winning family, selected parameters, validation scores, test metrics, typical values and category levels.

Table 4.2 Project artifacts and their analytical role

Artifact	Role
config.json	Business context, target and feature schema
dataset.csv	Cleaned labelled project snapshots
pipeline.joblib	Fitted preprocessing and classifier
metadata.json	Experiment results and explanation baselines
segmentation.json	Historical numeric customer profiles
plots/	Evaluation and exploratory graphics

Table 4.2 describes model and dataset organisation, not a multi-user isolation or access-control guarantee. The active selection is stored centrally, and the application is best understood as a local analytical prototype. Deploying it for simultaneous users requires an explicit session, storage and permission design. Serialized model files should be loaded only from trusted sources because deserialisation is not a safe way to inspect an unknown file.

4.3 Preprocessing Pipeline

The selected predictors are split into numeric and categorical branches within a ColumnTransformer. The numeric branch applies median imputation followed by standard scaling. The categorical branch applies most-frequent imputation followed by one-hot encoding, with the first level dropped and unknown categories ignored. The transformed branches are combined before the classifier is fitted.

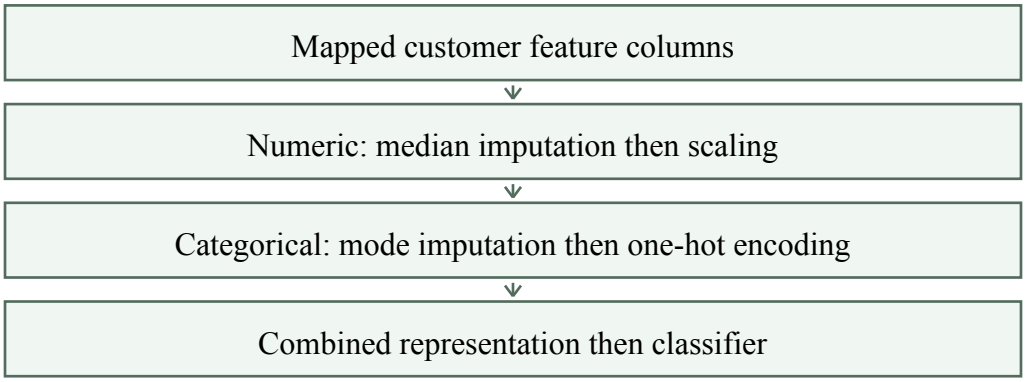


Figure 4.2 Preprocessing and classification pipeline

A pipeline is passed to GridSearchCV rather than fitting transformations on the complete dataset beforehand. Consequently, fold-specific medians, scaling statistics and category encodings are learned from the fold training records. This follows the fitting-boundary guidance in [7]. Unknown categories can be transformed without a runtime error, but that property does not mean the model has learned their business meaning. Missingness patterns and new categories should still be monitored.

4.4 Partitioning And Model Selection

The training code constructs an 80/20 stratified train-test split using random_state = 42. Within the training partition, five stratified folds are shuffled using the same seed. The search score is the area under the receiver operating characteristic curve. Each candidate family is searched over its configured parameter grid and refitted on the training partition. The family with the highest stored mean validation score is selected.

Table 4.3 Implemented hyperparameter search grids

Family	Parameter values	Count
Logistic regression	C: 0.1, 1.0, 10.0	3
Random forest	Trees: 50, 100; depth: 5, 10	4
Gradient boosting	Trees: 50, 100; rate: 0.05, 0.1; depth: 3, 5	8

There are fifteen parameter combinations across the three families, each evaluated over five folds. The code stores mean scores to four decimal places and compares those stored means across families. The selected pipeline is fitted on the complete training partition before test evaluation. Reported cross-validation values in this document come from the supplied metadata; a new complete search was not performed for this report.

4.5 Classification Metrics

A confusion matrix records true negatives, false positives, false negatives and true positives for a chosen classification threshold. Accuracy measures the fraction of all correctly classified records. Precision measures the fraction of predicted positive records that are positive. Recall measures the fraction of positive records detected. The harmonic mean of precision and recall is the F1 score.

Table 4.4 Definitions used for classification metrics

Metric	Expression
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1	2 x precision x recall / (precision + recall)
False-positive rate	FP / (FP + TN)

The receiver operating characteristic curve compares true-positive and false-positive rates across score thresholds. Its area assesses discrimination rather than the correctness of a single operating threshold. The held-out confusion matrices in Chapter 6 correspond to classifier predictions at the default binary decision rule. They must not be confused with the interface risk bands, whose upper cutoff is 0.60 in individual views and 0.65 in batch scoring, which are presentation categories rather than a validated business-cost policy.

4.6 Global And Local Model Interpretation

Global permutation importance is calculated on the held-out partition with ten repetitions, using the receiver-operating-characteristic area as the score. A positive mean decrease suggests that shuffling the feature harmed this fitted model's ranking performance. A value near zero does not prove the feature has no business relevance. Correlation, interactions, sample variation and the fitted model all affect the result [8].

Local sensitivity begins with the estimated probability for the customer. The code first fills omitted inputs from saved training baselines. For telecom inputs it derives omitted TotalCharges as tenure multiplied by MonthlyCharges. It substitutes a feature baseline and subtracts the resulting probability from the original. When tenure or MonthlyCharges is perturbed, TotalCharges is also recomputed, so those sensitivities are linked-input changes rather than strictly isolated one-feature effects. Positive differences are displayed as risk-increasing associations relative to that baseline; negative differences indicate the opposite direction within the model.

Table 4.5 Interpretation methods and boundaries

Method	What it supports	What it does not prove
Permutation diagnostic	Model reliance on shuffled features	Causal business importance
Baseline perturbation	Local probability sensitivity	Additive attribution identity
Review heuristic	Possible topic for analyst review	Intervention effectiveness

The local differences need not sum to the difference from a complete baseline customer. They are not Shapley values or causal treatment effects. The accompanying suggestions are rule-based review considerations, not learned recommendations with measured retention uplift.

4.7 Customer Segmentation Method

Segmentation is a separate exploratory process. The implementation takes numeric customer features, imputes missing values by the median and scales them before fitting K-means. The default request is four groups, with the actual count restricted by dataset size. The routine currently declines to produce segments when fewer than two numeric features or fewer than thirty rows are available. These are design choices of this application, not mathematical impossibility statements about one-dimensional clustering.

K-means minimises within-group squared distance in the transformed feature space [6]. After assignment, the program calculates cluster sizes, feature medians and observed churn rates. Persona labels compare group medians with global medians and apply rule-based wording. Churn is used for the subsequent profile summary, not as an input dimension of the clustering fit.

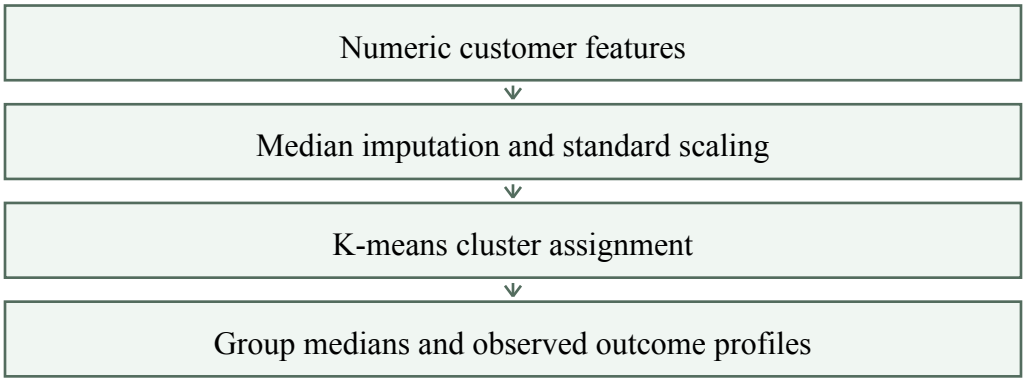


Figure 4.3 Exploratory segmentation workflow

These profiles are descriptive summaries of the analysed dataset. They are not held-out cluster generalisation scores and do not establish that cluster membership causes churn. New-customer segment assignment would require preserving and reusing the complete preprocessing-plus-clustering transformation.

4.8 Decision Support And Responsible Use

The final analytical step is a human review. The interface shows probabilities, risk bands, local sensitivities and historical distributions, but a business must decide which cases merit contact. A model score should be considered alongside contact consent, service history, customer needs and the cost of an unnecessary intervention. A low score should not remove access to normal service or support.

Table 4.6 Operational questions before real deployment

Question	Required action
Was each predictor known in time?	Audit feature measurement and label timing
Does the split match use?	Add time-ordered or prospective evaluation
Are probabilities calibrated?	Assess reliability on suitable validation data
Are groups affected differently?	Review errors and fairness by relevant groups
Does outreach help?	Run a controlled and approved intervention study

Table 4.6 identifies deployment evidence beyond the present prototype. Statistical fitting boundaries reduce preprocessing leakage, but a post-cancellation attribute can still leak the answer semantically. Human confirmation and data documentation are therefore necessary. Predictive association, customer segmentation and causal business impact remain separate claims throughout this report.

CHAPTER 5
IMPLEMENTATION

5.1 Implementation Environment

The current revision adds Windows and macOS launcher scripts, which were inspected but not executed on those operating systems. The application is implemented in Python with Flask for server-side routing, pandas for data handling, scikit-learn for modelling, joblib for persistence and Matplotlib for saved charts. HTML templates present the wizard, scoring and analytical views. The evaluated environment used Python 3.11 with the supplied exact package versions. The scikit-learn 1.8.0 dependency requires Python 3.11 or later [9].

Table 5.1 Runtime package versions in the supplied requirements

Package	Pinned version
pandas	2.3.3
scikit-learn	1.8.0
Flask	3.1.2
joblib	1.5.3
Matplotlib	3.10.7

The development requirements include pytest 9.1.1, ReportLab 4.4.10 and PyMuPDF1.26.7. The test execution described in Chapter 6 used the supplied runtime and development pins. No measured minimum processor speed, memory requirement or universal operating-system compatibility is asserted. Training time depends on dataset size, transformed feature count, grid size and the available machine.

5.2 Wizard: Data And Business Profile

The current home page links to the wizard, individual and batch scoring, and a live probability simulator. The wizard data-entry screen asks the user to upload a customer file and describe the project context. It also provides the two bundled demonstration datasets. The business profile includes a project name, business label, churn definition and observation-window text. These fields document intended meaning; the system does not infer a valid prediction horizon merely from that text.

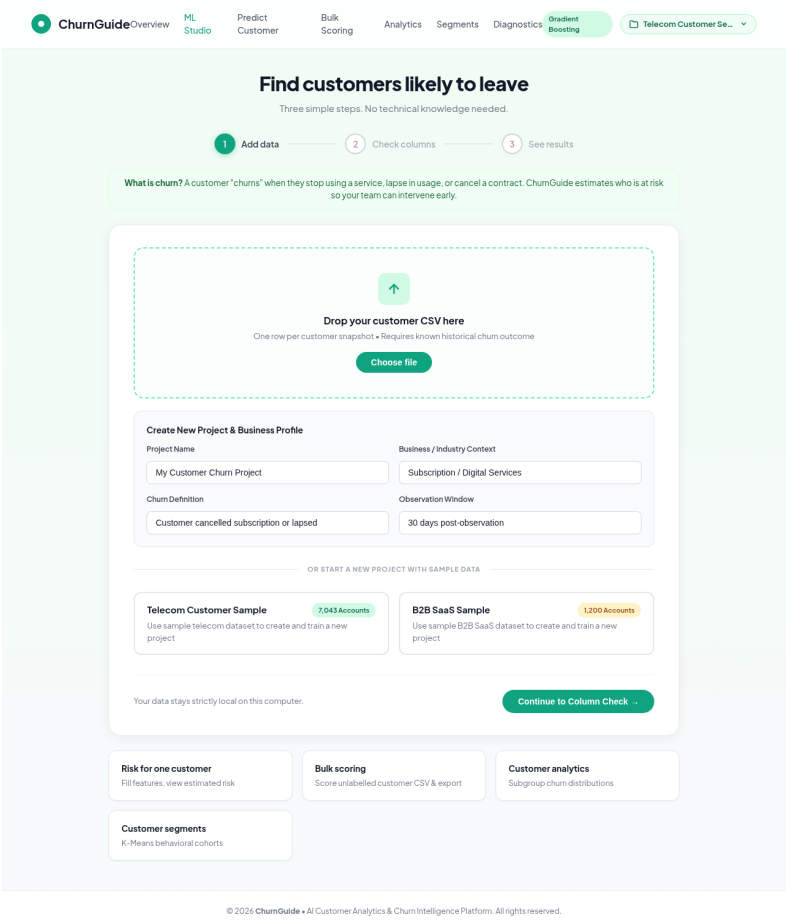


Figure 5.1 Data-selection and business-profile screen from the supplied application

Figure 5.1 shows the implemented three-step interface. A user should prepare a labelled historical training file for this step. An unlabelled current-customer table belongs in batch scoring after a project has been trained.

5.3 Wizard: Column Review And Results

Column review presents inferred numeric and categorical types, sample values, missingness and candidate identifier or target columns. A user selects the outcome, its positive value and the usable predictors. Type inference is an aid, not semantic verification. A binary demographic field, for example, is not automatically a churn outcome merely because it contains two values.

Table 5.2 Main wizard stages and expected user decisions

Stage	User decision	Result
Add data	Choose table and business context	Staged input and profile
Check columns	Confirm target, positive value and predictors	Validated project schema
See results	Inspect fitted model and customer sample	Links to scoring and analysis

The result screen summarises the current project and includes a ranked preview generated from the first one hundred dataset rows. The displayed top ten are therefore top-ranked within that sample, not a complete-portfolio top-ten ranking. For wider review, batch scoring can be used on the complete customer table. This distinction avoids confusing an interface sample with an exhaustive customer-priority list.

The project record is saved after successful validation and training. Its feature list then drives the fields and transformations used by later screens. Analysts should retain their original source data and record the meaning and timing of every included attribute.

5.4 Individual Customer Scoring

The individual-scoring form is generated from the active project's numeric features and saved categorical levels. On submission, the fitted pipeline returns an estimated positive-class probability. The explanation module adds a risk band, a typical-customer comparison and baseline-perturbation sensitivity values. This makes the form adapt to the active schema rather than requiring a telecom-specific layout.

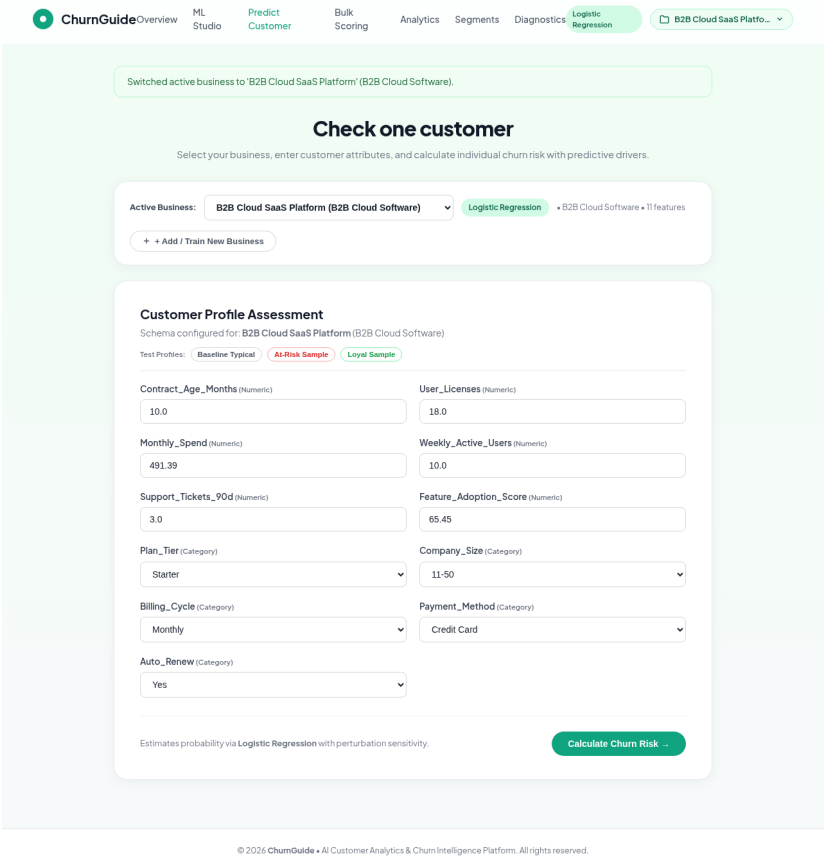


Figure 5.2 Software-account individual-scoring screen; the underlying demo data is synthetic

Figure 5.2 illustrates a different set of fields from the telecommunications project. The view demonstrates schema flexibility. It does not establish real software-company predictive accuracy or the effectiveness of suggested account-review actions.

5.5 Batch Scoring And Customer Analytics

Batch scoring uses a trained project and an unlabelled table containing the required predictor columns. The scoring function normalises header whitespace and case for matching and applies the saved pipeline. It appends Predicted_Churn_Probability and Risk_Level. The browser shows a limited preview and offers a scored file for download. The label is not required at inference time.

Table 5.3 Display thresholds by scoring view

View	Low	Medium	High
Individual / simulator	p < 0.35	0.35 <= p < 0.60	p >= 0.60
Batch output	p < 0.35	0.35 <= p < 0.65	p >= 0.65

The descriptive analytics page computes historical categorical subgroup rates and numeric summaries from the stored project dataset. It reports counts, distributions and contrasts between recorded outcomes. These are observations about the analysed snapshot, not a fresh test of the classifier. A subgroup with a high recorded churn rate is an analytical lead for review, not proof that the category causes departure.

Table 5.3 states presentation thresholds. Thresholds should not be adopted as an outreach policy without validation against business costs and available review capacity. Scored outputs also require a clear association with the intended project and input table before they are used operationally.

5.6 Model And Segment Views

The model view exposes the candidate comparison, selected parameters, held-out scores, confusion matrix, receiver-operating-characteristic curve and permutation diagnostic. The segment view displays saved numeric-cluster profiles and their observed outcome rates. Both views are intended to make the analytical process inspectable rather than leave only a final customer score.

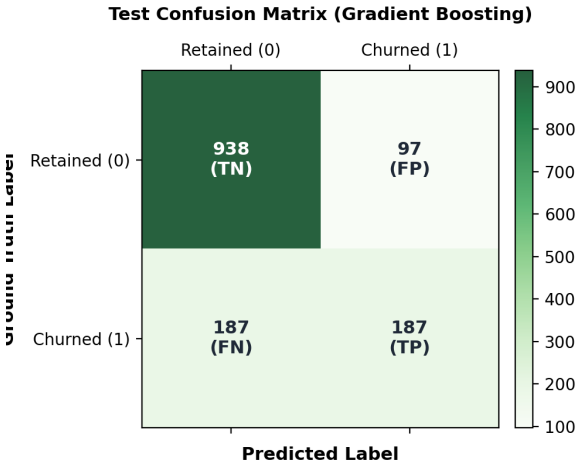


Figure 5.3 Held-out telecommunications confusion matrix saved by the supplied project

Figure 5.3 is a saved project artifact, not a new training experiment created for this report. Its four cell counts are discussed in Chapter 6. The complete model view also contains methodological text; the defensible scope is pipeline-based preprocessing and recorded split evaluation, not an absolute guarantee against all forms of leakage.

CHAPTER 6
RESULTS AND DISCUSSION

6.1 Evaluation Evidence And Reproducibility

Three evidence sources are kept separate. First, the supplied source defines the partitioning and parameter-search procedure. Second, the supplied project metadata and plots record the experiment outputs. Third, the review executed the included test suite and independently obtained held-out predictions from the saved classifiers using the declared split and cleaned data. The recomputed metrics matched the saved result values.

Table 6.1 Evidence available for this report

Evidence	Scope
Source inspection	Implemented training and analytical methods
Saved experiment records	Candidate scores, parameters and plots
17 of 18 tests passed	One home-navigation expectation differs
Saved-model recomputation	Prediction metrics on declared held-out rows
Browser checks	Observed principal desktop workflows

A complete fresh grid search was not run for preparation of the report. The cross-validation tables are therefore accurately identified as supplied experiment records. The independent check supports consistency of saved-model test predictions, not proof that every original training event occurred exactly once. No execution-time benchmark, calibration study, prospective field trial or retention-uplift experiment was performed.

6.2 Telecommunications Model Comparison

Table 6.2 Recorded five-fold training-partition comparison for telecommunications

Model	Mean area	Fold std.	Selected settings
Logistic regression	0.8463	0.0128	C=10.0
Random forest	0.8446	0.0115	Depth 5; trees 50
Gradient boosting	0.8493	0.0129	Rate 0.05; depth 3; trees 100

Table 6.2 reports the best configuration within each candidate family. Gradient boosting has the highest recorded mean validation area and is the selected telecommunications model. Its mean exceeds logistic regression by 0.0030. These closely spaced means and fold variations do not establish a statistically significant superiority claim. The search used the same training partition and scoring approach for all three families.

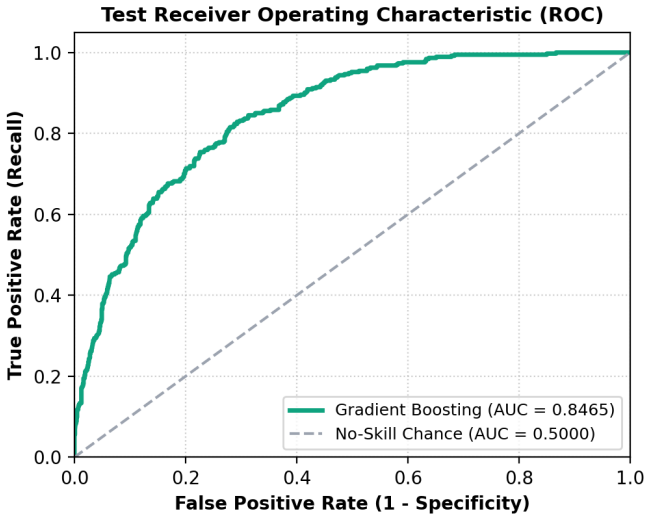


Figure 6.1 Saved telecommunications held-out receiver-operating-characteristic curve

Figure 6.1 displays the final selected model on the held-out partition. The reported area is 0.8465. The test result is used for reporting the chosen model, not for selecting another family after seeing this curve.

6.3 Telecommunications Held-Out Results

Table 6.3 Telecommunications held-out classification results

Measure	Value
Test records	1,409
Accuracy	0.7984
Precision	0.6585
Recall	0.5000
F1	0.5684
Area under curve	0.8465
TN / FP / FN / TP	938 /97 /187 /187

The matrix contains 1,035 retained customers and 374 churned customers. Of the churned customers,187 are detected and 187 are missed at the default decision rule. Of 284 positive predictions,187 are correct and 97 are false positives. These counts explain why ranking area can be reasonably strong while recall at a particular threshold is only one half.

An always-retained classifier would be correct on 1,035 of 1,409 test records, approximately 73.46 percent. The model's 79.84 percent accuracy is higher, but accuracy alone hides the missed positive cases. The combination of precision, recall and ranking area is therefore more informative than a single accuracy headline. Lowering a threshold may increase recall at the cost of additional false positives; the report does not invent results for a threshold sweep that was not performed.

6.4 Generated Software Demonstration Results

Table 6.4 Recorded training comparison on the synthetic software fixture

Model	Mean area	Fold std.	Selected settings
Logistic regression	0.7084	0.0390	C=0.1
Random forest	0.6996	0.0297	Depth 5; trees 50
Gradient boosting	0.6956	0.0269	Rate 0.05; depth 3; trees 50

Table 6.5 Synthetic software held-out results

Measure	Value
Records	240
Accuracy / precision	0.6542 /0.6049
Recall / F 1	0.4900 /0.5414
Area under curve	0.6894
TN / FP / FN / TP	108 /32 /51 /49

Logistic regression is selected for this fixture. The result shows that the model-selection procedure need not choose the same family for a different table. It does not establish performance on real enterprise accounts. The held-out set contains 100 positive and 140 negative records; the classifier detects 49 positives and misses 51. All interpretation of Table 6.4 and Table 6.5 remains conditional on the synthetic generation process.

6.5 Feature And Segment Interpretation

Table 6.6 Top saved permutation diagnostics: mean decrease in held-out area

Dataset	Feature	Decrease
Telecom	tenure	0.07385
Telecom	Contract	0.04300
Telecom	InternetService	0.02886
Synthetic software	Contract_Age_Months	0.09839
Synthetic software	Billing_Cycle	0.04561
Synthetic software	Feature_Adoption_Score	0.02679

Table 6.6 indicates the largest saved mean score decreases among the selected features. The repeat spread represents the standard deviation over permutation repetitions. It is not a confidence interval or standard error of an estimated population effect. Feature importance describes dependence of this fitted model on the supplied data, not an independent causal contribution.

Table 6.7 Saved telecommunications exploratory groups

Group	Customers	Share	Observed churn
4	2,276	32.3%	48.2%
3	1,704	24.2%	24.6%
2	1,904	27.0%	15.4%
1	1,159	16.5%	5.0%

Table 6.7 describes all supplied telecommunications rows grouped by numeric clustering. Group 4 has a median tenure of 14 months and Group 1 a median of 53 months. These summaries are descriptive, not independent test-set evidence or causal explanations.

6.6 Software Validation And Discussion

On the latest supplied revision, seventeen of eighteen tests passed under Python 3.11. The remaining assertion expects the former home-page redirect; the new landing page returns 200 instead. Separate route checks returned normal responses for the principal pages. Two simulator requests returned valid probabilities. This evidence is limited to the tested inputs and does not establish complete interface acceptance.

Table 6.8 Validation categories and interpretation

Category	Observed evidence	Boundary
Data and pipeline	Schema and preprocessing assertions pass	Not semantic leakage proof
Saved models and scoring	Probability and batch assertions pass	Not calibration or multi-user validation
Navigation	New home 200; old test expects 302	Suite is not fully passing

The unchanged saved models and recorded test metrics support the reported analytical results. Cross-sectional data, a single split seed, unverified outcome timing and synthetic software-account data limit generalisation. No production readiness or measured reduction in churn is claimed.

CHAPTER 7
CONCLUSION AND FUTURE WORK

7.1 Conclusions

ChurnGuide integrates business-profile setup, schema mapping, pipeline-based model comparison, individual scoring, batch scoring, descriptive analytics and numeric customer segmentation. The supplied demonstration projects show that the interface and model artifacts can accommodate two different customer-table schemas. The trained models remain project-specific, and generated software-account data is identified as a fixture rather than real enterprise validation.

For the public telecommunications snapshot, the supplied gradient-boosting model records a held-out ranking area of 0.8465 and accuracy of 0.7984. Its recall is 0.5000 at the default decision rule, so a substantial share of positive records is missed. For the synthetic software fixture, the supplied logistic-regression model records a held-out area of 0.6894. These results were consistent with an independent prediction check using the saved pipelines and the declared split.

The modelling process keeps statistical preprocessing within the training pipeline and uses training-partition cross-validation to compare candidates. Model sensitivity and permutation diagnostics make the fitted behaviour more inspectable, while the report distinguishes those associations from causal effects. The project is consequently best understood as a local decision-support prototype with reproducible saved artifacts and a clear need for domain-specific validation before operational use.

7.2 Future Work

The first research direction is time-aware validation. A real deployment should define an observation cutoff, a future outcome window and a policy for customers with incomplete follow-up. Multiple temporal test periods would provide stronger evidence about changing customer populations than a single random split. Repeated validation and uncertainty estimates would also help distinguish small candidate-score differences.

The second direction is decision quality. Probability calibration, precision-recall analysis and cost-aware threshold selection should be evaluated using data reserved for that purpose. A controlled retention study is needed before claims about the benefit of outreach or incentives. Review suggestions can then be assessed against actual customer outcomes rather than accepted because they sound plausible.

The third direction is deployment engineering. Session-specific project state, explicit scoring/export context, strict input diagnostics, complete clustering-pipeline persistence and narrow-screen adaptation would strengthen operational use. A deployed service also requires authentication, access controls, secure storage, auditability and a documented data-retention policy. These additions are future work and are not represented as completed features.

Table 7.1 Future work and the evidence it would add

Area	Evidence to obtain
Temporal evaluation	Performance on later customer periods
Calibration and thresholds	Reliable probability and cost trade-offs
Retention experiment	Measured effect of approved interventions
Deployment validation	Multi-user, security and device acceptance

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APPENDIX A
DATA DICTIONARIES

A.1 Telecommunications Feature Dictionary

The following dictionary records the feature roles in the supplied public telecommunications project. Numeric typing follows the project configuration, including conversion of TotalCharges text to numeric values. Names are preserved exactly so a reader can relate the dictionary to config.json and dataset.csv.

Table A.1 Telecommunications project feature roles

Role	Exact column names
Numeric	tenure; MonthlyCharges; TotalCharges
Categorical	gender; SeniorCitizen; Partner; Dependents; PhoneService; MultipleLines; InternetService; OnlineSecurity; OnlineBackup; DeviceProtection; TechSupport; StreamingTV; StreamingMovies; Contract; PaperlessBilling; PaymentMethod
Excluded identifier	customerID
Mapped outcome	Churn; Yes becomes 1 and No becomes 0

Identifiers are used for record association, not prediction. A field name alone is not sufficient evidence of its measurement date or business meaning. The project configuration and public source should be consulted together when adapting the sample.

A.2 Generated Software Feature Dictionary

Table A.2 Synthetic software project feature roles

Role	Exact column names
Numeric	Contract_Age_Months; User_Licenses; Monthly_Spend; Weekly_Active_Users; Support_Tickets_90d; Feature_Adoption_Score
Categorical	Plan_Tier; Company_Size; Billing_Cycle; Payment_Method; Auto_Renew
Excluded identifier	Account_ID
Mapped outcome	Status; Churned is the positive value

The supplied generator is scripts/generate_saas_data.py. The fixture exists to test cross-schema behaviour and is explicitly synthetic. Account counts, model results and feature relationships in this project must therefore not be described as observations from a real software business.

For a new organisation, a data dictionary should add units, allowed values, missing-value conventions, source system, measurement cutoff and label follow-up. Those details are not created automatically by the application. They are required for a defensible operational interpretation of an estimated churn probability.

APPENDIX B
REPRODUCIBILITY AND USER WORKFLOW

B.1 Reproducibility Checklist

The supplied saved models were inspected under Python 3.11 with the exact dependency pins listed in Table 5.1. To reproduce the experiment procedure, preserve the source data, configuration and software versions. Make a separate working copy before running tests or training so that active-project state and generated artifacts do not overwrite the reference copy.

Table B.1 Reproduction steps and expected evidence

Step	Evidence to retain
1. Prepare environment	Python and package version list
2. Inspect mapping	Target, positive class, IDs and predictors
3. Recreate split	80/20 stratified; seed 42
4. Search on training rows	Five-fold scores and selected parameters
5. Evaluate selected model	Held-out probabilities and confusion matrix
6. Save artifacts	Pipeline, configuration, metadata and plots

For this report, the eighteen included tests were run on the latest archive: seventeen passed and the home-redirect assertion failed after the landing-page change. Saved-model test metrics were checked. A complete new search is not claimed. Exact reproducibility can also depend on library builds and numerical behaviour; fixed seeds are an aid, not a guarantee across arbitrary environments.

B.2 User Workflow And Interpretation Checklist

For a new training project, prepare a historical table with known outcomes, define churn and confirm the outcome window. Upload the file through the wizard, inspect the schema and select the intended target and positive value. Exclude identifiers and fields that would not be available before the outcome. Review missingness and feature types before starting training.

After training, inspect the selected model and its held-out results before reviewing customers. Use the individual form for a single record and batch scoring for an unlabelled portfolio with the required predictors. Interpret risk bands as display categories. Treat local feature differences as sensitivity relative to a typical training value, and treat review suggestions as heuristics.

Before using a scored file, verify the intended project, feature schema and customer identifiers. Keep the analysed input and resulting output together. Historical analytics and segment profiles describe the stored dataset; they do not independently validate future outcomes. Do not infer intervention effectiveness from a high-risk label or a descriptive subgroup difference.

Table B.2 Reading the main outputs

Output	Question to ask
Probability	Was this model validated for this population?
Sensitivity	Is the baseline substitution meaningful?
Segment profile	How large and stable is this group?
Test score	Does the split match the intended future use?